PUSHING A NORMALIZED DUALIZING COMPLEX FORWARD ALONG A FINITE LOCAL MAP STAYS NORMALIZED

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The statement that the pushforward of a dualizing complex along a finite map is dualizing. Invoked at the start of the counted proof (source lines 3145-3146). Gives the dualizing property of the pushforward.
- The adjunction identifying the internal hom over the target with the internal hom over the source. Invoked inside the counted proof (source lines 3147-3151). Computes the relevant internal hom.

Source reference. Original lemma lemma-dualizing-finite: sources/stacks-dualizing/pinned-original-source.tex, source lines 3025–3025. Navigation pointer only: the counted text refers to the Stacks item lemma-dualizing-finite, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Source reference. Original lemma lemma-right-adjoint: sources/stacks-dualizing/pinned-original-source.tex, source lines 2428–2428. Navigation pointer only: the counted text refers to the Stacks item lemma-right-adjoint, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let $(A, \mathfrak{m}, \kappa) \to (B, \mathfrak{m}', \kappa')$ be a finite local map of Noetherian local rings. Let $\omega_A^\bullet$ be a normalized dualizing complex. Then $\omega_B^\bullet = R\mathrm{Hom}(B, \omega_A^\bullet)$ is a normalized dualizing complex for B.*

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Proof. By Lemma lemma-dualizing-finite the complex $\omega_B^\bullet$ is dualizing for B. We have

$$R\mathrm{Hom}_B(\kappa', \omega_B^\bullet) = R\mathrm{Hom}_B(\kappa', R\mathrm{Hom}(B, \omega_A^\bullet)) = R\mathrm{Hom}_A(\kappa', \omega_A^\bullet)$$

by Lemma lemma-right-adjoint. Since κ' is isomorphic to a finite direct sum of copies of κ as an A-module and since $\omega_A^\bullet$ is normalized, we see that this complex only has cohomology placed in degree 0. Thus $\omega_B^\bullet$ is a normalized dualizing complex as well. $\square$